Frequency-Aware Bias Calibration Attention Network for Hyperspectral Image Classification

Zhenqiu Shu, *Member, IEEE*, Nanbo Liu, Kexin Zeng, Hongbin Wang, and Jun Zhou, *Fellow, IEEE*

***Abstract*—Recently, self-attention mechanisms have significantly advanced hyperspectral image classification (HSIC) by enabling effective modeling of long-range spatial–spectral dependencies. However, existing methods often suffer from blurred land-cover boundaries and insufficient discriminability between spectrally similar classes due to redundant global interactions and inadequate preservation of fine-grained spatial details. To address these challenges, we propose a novel frequency-aware bias calibration attention network (FBCANet) for HSIC. Specifically, we design a frequency-domain spatial–spectral global aggregation module to model global contextual dependencies in the frequency domain. Within this module, a global frequency enhancement filter (GFEF) explicitly decouples complex frequency representations to adaptively suppress redundant interference caused by similar spectral signatures across different categories. Furthermore, we introduce a bias-calibrated global attention (BCGA) module, which compensates for the representation bias introduced by global attention modeling. This module consists of a global feature aggregation backbone and an explicit bias-calibration branch that extracts fine-grained spatial–spectral cues and injects them into the global interaction process, thereby enhancing interclass discriminability. In addition, a dual-path frequency-aware interaction module is proposed to capture high-frequency information and dynamically compensate for the loss of detailed textures and complex land-cover boundaries. Extensive experiments on three benchmark hyperspectral datasets demonstrate that the proposed method learns highly discriminative feature representations and consistently outperforms state-of-the-art HSIC methods. The source code for this work will be publicly available at https://github.com/szq0816/FBCANet**

***Index Terms*—Bias-calibrated, feature aggregation, fine-grained textures, frequency-domain, high-frequency, hyperspectral image classification (HSIC), spectral–spatial modeling.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture hundreds of contiguous spectral bands, providing rich spectral-material structures and long-range spectral continuity that are essential for accurate material discrimination. Owing to these distinctive physical characteristics, HSIs have become indispensable in a wide range of practical applications, including precision agriculture [1], environmental monitoring [2], and resource exploration [3]. As a fundamental task in these fields, HSI classification (HSIC) aims to assign a unique semantic label to each pixel. However, learning highly discriminative representations remains challenging due to inherent high-dimensional redundancy and complex coupling of spatial–spectral ambiguities. In particular, severe interclass spectral confusion and intricate boundary mixing substantially further hinder robust feature learning.

Over the past decade, deep learning [4], [5], [6], [7] has progressively outperformed traditional machine learning approaches [8], [9], [10] in HSIC tasks. Convolutional neural networks (CNNs) have achieved remarkable success in hierarchical feature extraction [11], [12]. For instance, 3-D convolutions [13], [14] and residual connections [15] have demonstrated remarkable capability in capturing complex joint spatial–spectral representations within local neighborhoods through structural designs. However, real-world HSIs also exhibit nonlocal material consistency and long-range spectral dependencies that require global receptive fields. Limited by the local receptive fields of fixed convolution kernels, conventional CNNs struggle to capture these essential long-range dependencies. Therefore, attention mechanisms have been extensively utilized in HSIC tasks in recent years [16], [17], [18]. They can dynamically model long-range dependencies and contextual correlations among tokens, yielding remarkable classification accuracy. However, attention inevitably mixes heterogeneous spectral responses, thus leading to feature oversmoothing. This redundant global aggregation not only amplifies interclass spectral ambiguity among similar materials, but also suppresses the high-frequency details crucial for delineating complex land-cover boundaries. Consequently, its discriminative capability is significantly compromised across various applications.

To mitigate these inherent limitations, several studies have developed hybrid architectures that integrate CNNs with attention mechanisms. For instance, Xu et al. [19] employed convolutional multihead attention units to capture long-range contextual dependencies while reducing computational overhead. Shu et al. [20] combined convolution-based local feature extraction with cross-attention-driven spatial–spectral interactions, enabling the collaborative modeling of fine-grained neighborhood structures and long-range contextual dependen-

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Zhenqiu Shu, Nanbo Liu, Kexin Zeng, and Hongbin Wang are with the Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming 650032, China (e-mail: shuzhenqiu@163.com; 2636458935@qq.com; 2240329175@qq.com; whbin2007@126.com).

Jun Zhou is with the School of Information and Communication Technology, Griffith University, Nathan, QLD 4111, Australia (e-mail: jun.zhou@griffith.edu.au).

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cies. In addition, Zhao et al. [21] introduced convolutional transformer (CT) blocks, which integrate convolutional operations with attention-based contextual modeling to jointly capture local and global features of HSI patches. Although these hybrid strategies improve spatial–spectral representation learning, they still operate exclusively within the spatial–spectral domain. Consequently, such spatial–spectral interactions can only partially alleviate boundary degradation and spectral mixing among heterogeneous materials. In particular, multiscale spatial attention still processes mathematically coupled signals, where high-frequency structural boundaries and low-frequency redundant information remain entangled.

Therefore, the frequency-domain transformation of spatial–spectral features facilitates explicit distinction between semantic and structural mixed information. Without this explicit decoupling, redundant frequency components inevitably exacerbate interclass spectral confusion and progressively obscure crucial structural boundaries during global aggregation. Recently, several studies have explored various frequency transformation techniques for HSIC. For instance, Zhang et al. [22] proposed a novel wavelet frequency-domain enhancement network for HSIC. It introduces a channel-shift spatial-awareness module and a wavelet frequency-domain feature enhancement module to enhance both texture and semantic representations. Liu et al. [23] proposed a frequency-enhanced spatial–spectral network that leverages frequency information to enhance feature representation and enable effective cross-modal interactions. To mitigate cross-domain discrepancies, Zhang et al. [24] designed a 3-D frequency-domain transform network that facilitates intradomain and interdomain information exchange while enabling spatial-frequency collaboration to learn representative frequency-domain features. Although these methods demonstrate the effectiveness of frequency-domain transformations, they primarily utilize frequency information as an auxiliary enhancement strategy rather than an explicit disentanglement framework. Therefore, these frequency enhancement strategies inevitably suffer from both redundant interference and informative structural signals, leading to spectral confusion and boundary degradation in complex surface coverage scenes.

To address these issues, we propose a novel frequency-aware bias calibration attention network (FBCANet) for HSIC. First, to alleviate interclass spectral confusion, the proposed FBCANet method projects features into the frequency domain to adaptively suppress redundant global interference while enhancing discriminative representations. Second, we introduce an explicit mechanism that extracts localized confidence maps and dynamically injects them as content-aware inductive biases into the global attention process. Unlike the static fixed-weight inductive biases of traditional CNNs, this dynamic calibration explicitly guides the attention distribution to preserve fine-grained neighborhood structures. Meanwhile, a parallel frequency-aware pathway extracts high-frequency spatial details to preserve complex land-cover boundaries. By integrating this calibrated global context with high-frequency local structures, the proposed method can learn highly discriminative representations from HSIs. Extensive experiments on three benchmark datasets demonstrate the superiority of the proposed method.

The main contributions of this article are summarized as follows.

1) We propose a novel FBCANet that pioneers the integration of frequency-domain decoupling and inductive bias compensation for HSIC. This method adaptively suppresses redundant low-frequency spectral information while explicitly embedding localized spatial–spectral priors, thereby addressing the inherent homogenization of standard attention and fundamentally alleviating interclass spectral confusion.
2) We introduce a dual-path interaction mechanism to tackle spatial boundary blurring. Specifically, this mechanism focuses exclusively on extracting high-frequency spatial components to capture local structural details. This targeted spatial compensation dynamically recovers fine-grained textures and complex boundary structures, explicitly preventing edge degradation during deep global aggregation and significantly enhancing the discriminative capability of the proposed model.
3) Experimental results on several benchmark datasets demonstrate that the proposed method consistently outperforms several state-of-the-art approaches in HSIC tasks.

The remainder of this article is organized as follows. Section II reviews the related work. Section III introduces the proposed method. Section IV presents experimental results. Finally, Section V concludes this work.

II. Related Work

In this section, we introduce HSIC methods based on attention mechanisms and frequency-domain modeling.

A. Attention-Based HSIC

In the field of HSIC, attention mechanisms have progressively evolved from simple feature enhancement techniques to joint spatial–spectral representation modeling frameworks. For instance, Xu et al. [25] developed a dual-selective fusion Transformer (DSFormer) that captures multiscale contextual information through adaptive spatial–spectral feature selection and fusion. However, it may fail to preserve pixel-wise geometric relationships when modeling long-range dependencies, thus insufficiently capturing multidirectional spatial–spectral interactions. To address this limitation, Khan et al. [26] proposed a learnable positional encoding to enhance geometric and multidirectional interaction modeling. Similarly, Wang et al. [27] incorporated center-neighborhood self-similarity descriptors to improve spatial representations, thereby effectively capturing irrelevant spatial–spectral information and local positional information, as well as spatial correlations between pixels. In addition, Zhou et al. [28] proposed a dual-branch CT network (DCTN) that integrates local and global spatial–spectral features via an efficient interaction self-attention module. This design effectively mitigates the

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deficiency of CNNs in capturing long-range global spectral dependencies and remedies the inefficiency of Transformers in encoding fine-grained local spatial details. Lu et al. [29] employed a kernel-selected multiscale aggregation method to facilitate the effective alignment and deep integration of cross-scale and multilevel features. Cui and Zhang [30] incorporated a cross-scan Mamba to mitigate the quadratic scaling of traditional self-attention mechanisms. Shu et al. [31] proposed a spectral–spatial synergy-guided attention network (SSSGAN) that improves interclass feature discriminability by jointly leveraging multilevel spectral semantics and multiscale spatial details. Nevertheless, the aforementioned methods lack a frequency-domain perspective for enhancing feature representations, making them highly susceptible to redundant interference arising from similar spectral signatures across different classes during global context aggregation. Consequently, their capacity to preserve fine-grained structural details and maintain clear discriminative boundaries in complex scenes remains inherently limited.

B. Frequency-Domain-Based HSIC

In recent years, frequency-domain-based HSIC methods have garnered increasing attention. These approaches excel at explicitly decomposing spatial–spectral information into distinct frequency bands, thereby substantially improving HSI feature representation. For instance, Shi et al. [32] proposed a multiscale hierarchical convolution-aided Fourierformer (MHCFormer) to characterize local spatial dependencies while adequately leveraging token generation and mixing mechanisms. Similarly, Fan et al. [33] introduced a frequency and topology interaction network (FTIN) to model complex relationships of HSIs. By stacking frequency-topology cross-unit, this network jointly learns frequency and topological features across spectral bands, thereby improving classification performance. Liu et al. [34] proposed a collaborative network via multihead sparse and high-low frequency interaction (MSFI-CNet), thus achieving effective sparsification and enhanced high-low frequency learning. In addition, Wan et al. [35] presented a frequency spectral-channel fusion and cross-scale global aggregation network to efficiently process multiscale and multifrequency information. It leverages a frequency-domain spectral feature extraction module and the fast Fourier transform (FFT) to facilitate spectral-channel fusion and cross-scale global aggregation. Additionally, Chen et al. [36] developed an attention-guided frequency feature decomposition network, which employs a Transformer-based frequency feature decomposition module to capture distinct directional frequency components from multimodal remote sensing data.

However, existing frequency-domain HSIC methods still suffer from insufficient decoupling of complex frequency representations, which inherently limits their ability to suppress redundant frequency interference.

III. Method

In this section, we introduce the overall framework of the proposed FBCANet method in detail.

A. Overview

Fig. 1 illustrates the overall architecture of the proposed framework, which consists of three main components: a frequency domain spatial–spectral global aggregation (FDSSGA) module that integrates the global frequency enhancement filter (GFEF) and bias-calibrated global attention (BCGA) modules, a local spectral feature extraction (LSFE) module, and a DCT-gated local spatial feature extraction (DCTLSFE) module. Let $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ denote the input HSIs, which are partitioned into several patches, yielding $\mathbf{f}_{\text{patch}}$. The patch features are then fed into two parallel branches. On the spectral branch, a shallow convolution is first applied to project the features, which are then fed into the several spectral residual blocks (SpecResBlock) to extract local spectral details. In parallel, the spatial branch employs a shallow convolution followed by the DFAI module to capture fine-grained textures and high-frequency structures. Meanwhile, the FDSSGA module is adopted to perform spatial–spectral modeling from a frequency-domain perspective. Specifically, the GFEF module employs the Fourier transform to adaptively enhance discriminative features. The resulting features are then processed by the BCGA module, where the bias branch is injected into the main Q-K interaction pathway to perform feature calibration. The outputs of these two branches are fused by the dual-branch fusion (DBF) module using parallel decoupled pathways for adaptive feature fusion. Finally, a linear classifier is used to produce the final predictions.

B. Frequency Domain Spatial–Spectral Global Aggregation

As shown in Fig. 1, the FDSSGA module comprises the GFEF and BCGA modules. Specifically, the input features are first processed by the GFEF module, which adaptively enhances spatial–spectral representations in the frequency domain. The enhanced features are subsequently fed into the BCGA module for global contextual modeling. The module explicitly extracts spatial–spectral information as a bias compensation term and uses it to calibrate global representations, thereby significantly enhancing interclass discriminability. Detailed architectural descriptions of the GFEF and BCGA modules are provided in the following sections.

1) Global Frequency Enhancement Filter: As illustrated in Fig. 2, we design the GFEF module to learn a more robust global representation from a frequency perspective. Inspired by complex-valued representations in the frequency domain [37], this module explicitly decomposes the input features into real and imaginary components via the Fourier transform. Specifically, the spatial-frequency branch projects features into the frequency domain using the 2-D real FFT (RFFT). Real and imaginary components are processed via nonshared convolutional streams, enabling independent modulation of amplitude and phase. Meanwhile, a frequency modulation module is utilized to dynamically enhance discriminative frequency representations. Following the inverse transformation, the processed features obtain significantly strengthened global frequency components. In contrast, tailored to the continuous-signal characteristics of hyperspectral bands, the spectral frequency branch performs a 1-D RFFT along the channel

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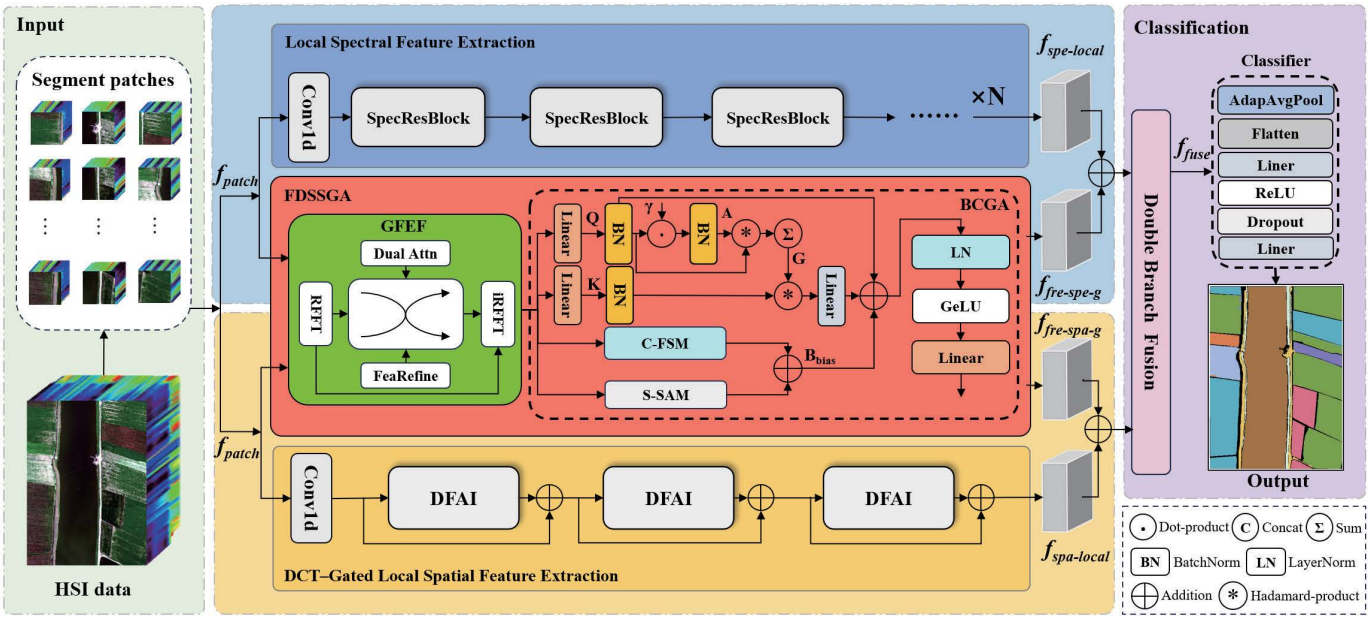


Fig. 1. Overall architecture of the proposed FBCANet method. It comprises three parallel feature extraction modules: a shared FDSSGA module that integrates GFEF and BCGA, an LSFE module, and a DCTLSFE module. Specifically, both the GFEF and BCGA modules within the FDSSGA module enhance the spatial and spectral feature representations and model global contextual relationships, respectively. The LSFE module uses a spectral residual module to extract local spectral features, whereas the DCTLSFE module incorporates the DFAI to capture fine-grained textures and high-frequency structures.

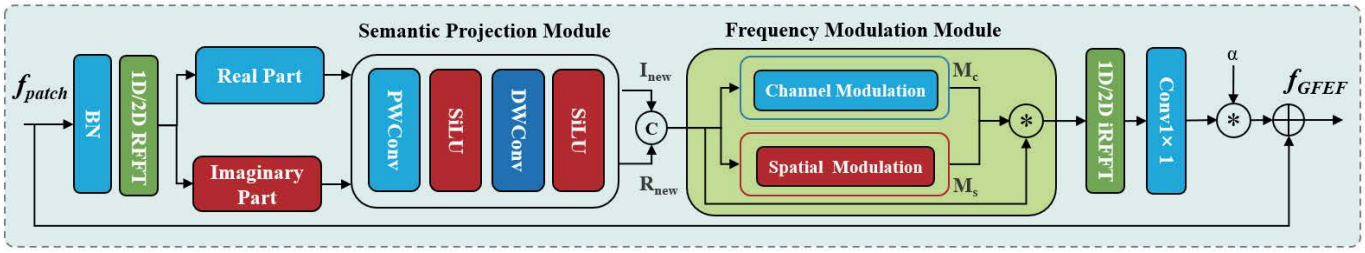


Fig. 2. Detailed architecture of the proposed GFEF module. The feature patches are projected into the frequency domain via the FFT and then decoupled into real and imaginary parts. Following a semantic projection process, a frequency modulation module is used to adaptively modulate the frequency-domain feature representations. It not only effectively suppresses redundant frequency information but also amplifies the critical discriminative frequency components.

dimension. In the frequency domain, it fine-tunes the amplitude and phase via a residual structure equipped with a learnable parameter α to aggregate features. This adaptively filters out redundant spectral information and amplifies critical absorption bands.

Specifically, given an input feature map $\mathbf{f}_{\text{patch}}$, we first stabilize the data distribution through batch normalization (BN). Subsequently, a 1-D/2-D RFFT is applied to convert the features into the frequency domain. Specifically, the spatial-frequency branch applies 2-D RFFT over the spatial dimensions ($H \times W$) and the spectral-frequency branch uses 1-D RFFT along the channel dimension to capture spectral variations. Let $\mathcal{F}(\cdot)$ denote the 1-D/2-D RFFT operation and i be the imaginary unit. The complex frequency-domain representation can be decoupled into a real part $\mathbf{R}$ and an imaginary part $\mathbf{I}$ as follows:

$$\mathcal{F}\left(\mathbf{f}_{\text{patch}}\right) = \mathbf{R} + i \cdot \mathbf{I}. \tag{1}$$

To capture intrinsic feature representations in the frequency domain, we design a semantic projection module that processes the real and imaginary components separately. This block consists of point-wise convolutions (PW-Conv) and depth-wise convolutions (DW-Conv), and then utilizes the SiLU activation function to enhance feature representation ability. Thus, we have

$$\mathbf{R}_{\text{new}} = \text{SiLU}(\text{DWConv}\left(\text{SiLU}\left(\text{PWConv}\left(\mathbf{R}\right)\right)\right)) \tag{2}$$

$$\mathbf{I}_{\text{new}} = \text{SiLU}(\text{DWConv}\left(\text{SiLU}\left(\text{PWConv}\left(\mathbf{I}\right)\right)\right)) \tag{3}$$

where $\text{SiLU}(\cdot)$ denotes the SiLU activation function. Subsequently, the processed real $\mathbf{R}_{\text{new}}$ and imaginary parts $\mathbf{I}_{\text{new}}$ are concatenated to obtain the fused frequency-domain features $\mathbf{f}_{\text{cat}}$ as follows:

$$\mathbf{f}_{\text{cat}} = \left[\mathbf{R}_{\text{new}}; \mathbf{I}_{\text{new}}\right]. \tag{4}$$

Although $\mathbf{R}_{\text{new}}$ and $\mathbf{I}_{\text{new}}$ originate from the complex frequency-domain representation, we treat them as complementary frequency feature components rather than reconstructing them into complex-valued signals. By concatenating the refined real and imaginary components, we can preserve the complementary frequency information encoded in both components and enable subsequent layers to adaptively learn their correlations. Notably, the concatenation operation expands the

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channel dimension from **C** to **2C**, where the first and second halves correspond to $\mathbf{R}_{\text{new}}$ and $\mathbf{I}_{\text{new}}$, respectively.

The frequency modulation module is applied to the concatenated features, thus further emphasizing salient frequency components. The channel map $\mathbf{M}_c$ compresses spatial dimensions via average pooling and models interchannel dependencies. Meanwhile, the spatial weight map $\mathbf{M}_s$ is generated via a convolutional operation. These two branches jointly refine the frequency-domain features as follows:

$$\mathbf{M}_c = \sigma\left(\text{MLP}\left(\text{AvgPool}\left(\mathbf{f}_{\text{cat}}\right)\right)\right) \tag{5}$$

$$\mathbf{M}_s = \sigma\left(\text{Conv}_{3\times 3}\left(\mathbf{f}_{\text{cat}}\right)\right) \tag{6}$$

$$\mathbf{f}'_{\text{freq}} = \mathbf{f}_{\text{cat}} * \mathbf{M}_c * \mathbf{M}_s \tag{7}$$

where σ denotes the Sigmoid function, $*$ represents Hadamard product, and $\odot$ denotes the dot product. The enhanced frequency features $\mathbf{f}'_{\text{freq}}$ are decoupled back into the updated real part $\mathbf{R}'$ and imaginary part $\mathbf{I}'$. After reconstructing the complex representation, a 1-D/2-D inverse RFFT (IRFFT) is applied to map the features back to the spectral and spatial domains, respectively. Finally, we apply a convolutional layer for projection, and construct a residual connection with the original input weighted by a learnable weight α as follows:

$$\mathbf{f}_{\text{GFEF}} = \alpha * \text{Conv}\left(\mathcal{F}^{-1}\left(\mathbf{R}' + i \cdot \mathbf{I}'\right)\right) + \mathbf{f}_{\text{patch}} \tag{8}$$

where $\mathcal{F}^{-1}(\cdot)$ denotes the IRFFT operation. The learnable weight α aims to adaptively control the contribution of the frequency-enhanced features to the final output $\mathbf{f}_{\text{GFEF}}$. This incorporates rich frequency-domain details while preserving the original structure.

2) Bias-Calibrated Global Attention: Inspired by [38] and [39], we design the BCGA module to overcome the limitations of traditional self-attention in distinguishing spectrally similar classes. This module reduces the memory-intensive cost of $\mathbf{QK}^{\mathbf{T}}$ matrix multiplication operations. Instead, it achieves global contextual interactions by generating a pooled global query vector via learnable attention weights, which then interacts with the key matrix through element-wise multiplications. Specifically, given the input features $\mathbf{X} \in \mathbb{R}^{N\times C}$ (where $N = H \times W$ denotes the number of spatial tokens and C is the channel dimension), they are first mapped into $\mathbf{Q} \in \mathbb{R}^{N\times D}$ and $\mathbf{K} \in \mathbb{R}^{N\times D}$ representations, where D represents the projected embedding dimension. Unlike conventional attention mechanisms that rely on computing an exhaustive $N \times N$ similarity matrix, the BCGA module introduces a learnable structured global weight vector γ. By computing the dot product between $\mathbf{Q}$ and γ, we obtain an attention distribution that quantifies the global importance of each token. This distribution is then utilized to adaptively aggregate $\mathbf{Q}$ into a unified global context representation. Finally, the BCGA module performs Hadamard products to modulate the token-wise representations in $\mathbf{K}$ using the broadcasted global context. This enables effective bias-driven calibration while alleviating feature homogenization caused by redundant interactions among spectrally similar materials, making it particularly suitable for HSIs with high-dimensional spectral redundancy. Thus, we have

$$\mathbf{A}_{\text{spe}} = \mathcal{N}_{\text{spe}}\left(\mathcal{N}_{\text{spe}}\left(\mathbf{Q}\right) \odot \gamma\right) \tag{9}$$

$$\mathbf{A}_{\text{spa}} = \mathcal{N}_{\text{spa}}\left(\mathcal{N}_{\text{spa}}\left(\mathbf{Q}\right) \odot \gamma\right) \tag{10}$$

where $\mathcal{N}_{\text{spe}}$ and $\mathcal{N}_{\text{spa}}$ denote the BN applied to the spectral and spatial tokens, respectively. The two normalization operations in (9) and (10) serve distinct roles. The first normalization stabilizes the scale of query features, and the second recalibrates the feature distribution following modulation with the learnable factor γ. This dual normalization strategy ensures consistent feature statistics, thereby improving the robustness of additive attention during global aggregation. This distribution is then used to adaptively aggregate $\mathbf{Q}$ into a unified global context representation $\mathbf{G}$ as follows:

$$\mathbf{G} = \sum_{i=1}^{N} \mathbf{A}_i * \mathbf{Q}_i. \tag{11}$$

The global context vector $G \in \mathbb{R}^{1\times D}$ is broadcast along the spatial dimension and then element-wise multiplied with $K \in \mathbb{R}^{N\times D}$ to achieve global context-guided token modulation.

Additionally, we introduce a parallel bias-correction branch to compensate for the inherent loss of local inductive biases. Specifically, two submodules, C-FSM and the S-SAM, are embedded to extract localized spatial and spectral priors. The C-FSM serves as a semantic filter to capture global discriminative spectral feature information. It performs global average pooling followed by a 1×1 convolution and a ReLU activation function to estimate channel-wise saliency, thereby suppressing redundant spectral bands. Meanwhile, the S-SAM submodule focuses on fine-grained spatial structure alignment by applying a 3×3 spatial convolution to batch-normalized features, thereby preserving critical land-cover boundaries. Notably, BN in S-SAM stabilizes the spatial feature distribution, and global statistical aggregation in C-FSM reduces spatial variation interference. These complementary operations align feature statistics and enable stable fusion of the two branches without additional learnable weighting. To construct the interactive bias, the spatial–spectral bias $\mathbf{B}_{\text{bias}}$ of both branches is combined in an additive manner as follows:

$$\mathbf{B}_{\text{bias}} = \text{ReLU}\left(\text{S-SAM}\left(\mathbf{X}\right)\right) + \text{ReLU}\left(\text{C-FSM}\left(\mathbf{X}\right)\right) \tag{12}$$

where ReLU$(\cdot)$ represents the ReLU activation function. This hybrid additive bias precisely regulates the allocation of contextual information across diverse land-cover textures and spectral bands, thereby effectively mitigating interference arising from similar spectral signatures across different classes. Subsequently, the BCGA module integrates these multidimensional perceptual descriptors. It dynamically modulates the $\mathbf{K}$ representations with $\mathbf{G}$ through the Hadamard product, formulated as follows:

$$\mathbf{Z} = \text{Proj}\left(\mathbf{G} * \mathbf{K}\right). \tag{13}$$

Here, Proj$(\cdot)$ denotes a linear projection layer. The modulated representation is then integrated with the original $\mathbf{Q}$ and the $\mathbf{B}_{\text{bias}}$ via residual connections. The final enhanced output $\mathbf{X}_{\text{out}}$ is formulated as follows:

$$\mathbf{X}_{\text{out}} = \text{Linear}\left(\text{GELU}\left(\text{LN}\left(\mathbf{Z} + \mathbf{Q} + \mathbf{B}_{\text{bias}}\right)\right)\right) \tag{14}$$

where LN$(\cdot)$ denotes layer normalization and Linear$(\cdot)$ represents a linear transformation.

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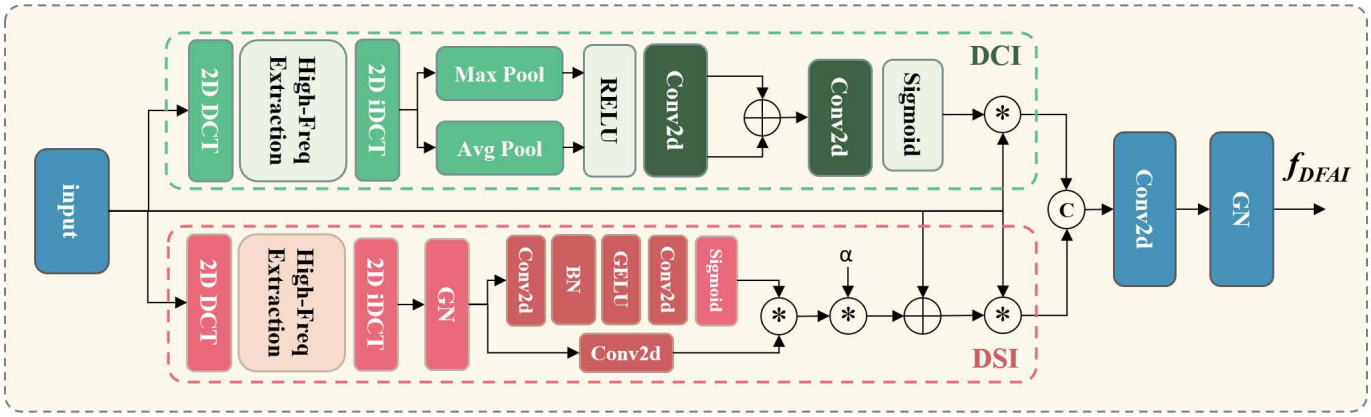


Fig. 3. Structure of the DFAI module. This module adopts a dual-pathway architecture to extract fine-grained spatial features. By suppressing low-frequency components, it effectively emphasizes salient spatial details and texture information, which are further refined through a gated interaction mechanism to enhance discriminative representations.

The BCGA module aggregates global contextual information with high precision and adaptively calibrates deep feature representations through a spatial–spectral bias module, thereby significantly enhancing discriminability between spectrally similar classes.

C. Local Spectral Feature Extraction

As shown in Fig. 1, the LSFE module comprises a convolutional layer and a stack of SpecResBlock. Each SpecResBlock includes two convolutional layers, two BN layers, and a ReLU activation function. Specifically, the partitioned input features $\mathbf{f}_{\text{patch}}$ are first processed by the initial convolutional layer to yield intermediate features $\mathbf{f}_{\text{conv}}$. The stacked residual modules subsequently refine these features to generate the local spectral representations.

The detailed mathematical formulation of each SpecResBlock is given as follows:

$$\text{SpecRes}(\mathbf{f}) = \text{BN}(\text{Conv}(\text{ReLU}(\text{BN}(\text{Conv}(\mathbf{f}))))) + \mathbf{f} \quad (15)$$

where $\mathbf{f}$ denotes the input features of each SpecResBlock, and SpecRes() denotes the output of each SpecResBlock module. Accordingly, the local spectral representations $\mathbf{f}_{\text{spe-local}}$ of the stacked residual network can be formulated as follows:

$$\mathbf{f}_{\text{spe-local}} = \text{SpecRes}\left(\cdots\left(\text{SpecRes}\left(\mathbf{f}_{\text{conv}}\right)\right)\right). \quad (16)$$

D. DCT-Gated Local Spatial Feature Extraction

As shown in Fig. 1, the DCTLSFE module is designed to extract spatial texture details. It consists of a convolutional layer and three residually stacked DFAI modules. This local spatial feature $\mathbf{f}_{\text{spa-local}}$ extraction process is mathematically formulated as follows:

$$\mathbf{Y}^{(0)} = \mathbf{f}_{\text{conv}} \quad (17)$$

$$\mathbf{Y}^{(l)} = \mathbf{Y}^{(l-1)} + \text{DFAI}_l\left(\mathbf{Y}^{(l-1)}\right), \quad l = 1, 2, 3 \quad (18)$$

$$\mathbf{f}_{\text{spa-local}} = \mathbf{Y}^{(3)} \quad (19)$$

where $\mathbf{f}_{\text{conv}}$ denotes the output of the convolutional operation, $\mathbf{Y}^{(l)}$ denotes the intermediate variable at the lth stage, and DFAI(·) represents the DFAI module. The structure of the DFAI module is shown in Fig. 3. This dual-pathway architecture comprises DCT spatial interaction (DSI) and DCT channel interaction (DCI), which are designed to capture local high-frequency information and discriminative edge information in a multidimensional manner. Specifically, the input features are processed in parallel through these two branches, yielding spatial-gated high-frequency residuals and channel attention-modulated representations, respectively. Finally, the outputs from both pathways are aggregated via element-wise addition, followed by a convolutional layer and group normalization (GN), yielding the final integrated output $\mathbf{f}_{\text{DFAI}}$.

The DSI pathway utilizes a frequency-domain residual to extract fine-grained structural details commonly over-smoothed by standard spatial convolutions. Specifically, it projects the features into the frequency domain using 2D-DCT, and then applies a high-pass filter weight mask $\mathbf{M}_h$ to explicitly preserve the high-frequency component $\mathbf{f}_{\text{high}}$ as follows:

$$\mathbf{f}_{\text{high}} = \text{DCT}_{2D}^{-1}\left(\mathbf{M}_h * \text{DCT}_{2D}\left(\mathbf{f}_{\text{conv}}\right)\right) \quad (20)$$

where $\mathbf{M_h} \in \mathbb{R}^{H\times W}$ denotes the binary high-pass mask. Since 2D-DCT concentrates low-frequency components in the top-left region, $\mathbf{M_h}$ suppresses these coefficients as follows:

$$\mathbf{M_h}(u, v) = \begin{cases} 0, & \text{if } u \le \lfloor H \cdot r \rfloor \text{ and } v \le \lfloor W \cdot r \rfloor \\ 1, & \text{otherwise} \end{cases} \quad (21)$$

where (u, v) represents frequency coordinates, and r denotes the cutoff ratio. Following frequency-domain modulation strategies [40], r is empirically set to 0.25.

We introduce a spatial gating network to dynamically filter out redundant frequency components and retain only semantically meaningful boundaries. Therefore, we can generate a spatial gate $\mathbf{G}_s$ as follows:

$$\mathbf{G}_s = \sigma\left(\text{Conv}\left(\text{GELU}\left(\text{BN}\left(\text{Conv}\left(\text{GN}\left(\mathbf{f}_{\text{high}}\right)\right)\right)\right)\right)\right). \quad (22)$$

The refined high-frequency features are then injected back into the main pathway via a learnable scaling parameter α

$$\mathbf{f}_{\text{DSI}} = \mathbf{f}_{\text{conv}} \cdot \left(\mathbf{f}_{\text{conv}} + \alpha * \left(\text{Conv}_{1\times 1}\left(\text{GN}\left(\mathbf{f}_{\text{high}}\right) * \mathbf{G}_s\right)\right)\right). \quad (23)$$

To ensure stable training, α is initialized to zero such that the module initially acts as an identity mapping. It is then

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optimized jointly with other network parameters under the same learning rate schedule. This gating mechanism allows the features $\mathbf{f}_{\text{DSI}}$ to enhance genuine geometric contours and suppress redundant information.

In a complementary fashion, the DCI pathway utilizes the high-frequency component to calibrate interchannel responses. It first extracts the high-frequency components $\mathbf{f}_{\text{high}}$ in a similar manner using 2D-DCT. To comprehensively capture the global features of each channel in these high-frequency maps, both adaptive max pooling ($\mathbf{S}_{\text{max}}$) and adaptive average pooling ($\mathbf{S}_{\text{avg}}$) are applied spatially as follows:

$$\mathbf{S}_{\text{max}} = \text{MaxPool}\left(\mathbf{f}_{\text{high}}\right) \tag{24}$$

$$\mathbf{S}_{\text{avg}} = \text{AvgPool}\left(\mathbf{f}_{\text{high}}\right). \tag{25}$$

Subsequently, grouped convolutions and the ReLU activation function are employed to process these descriptors independently, which are then aggregated to generate the adaptive channel weight vectors $\mathbf{W}_c$ as follows:

$$\mathbf{W}_c = \text{Conv}\left(\sigma\left(\text{Conv}\left(\text{ReLU}\left(\mathbf{S}_{\text{max}}\right) + \text{ReLU}\left(\mathbf{S}_{\text{avg}}\right)\right)\right)\right). \tag{26}$$

The original features are then modulated via a channel-wise Hadamard product, thus obtaining the feature representations $\mathbf{f}_{\text{DCI}}$ as follows:

$$\mathbf{f}_{\text{DCI}} = \mathbf{W}_c * \mathbf{f}_{\text{conv}}. \tag{27}$$

Furthermore, the DFAI module fuses the outputs from both the spatial and channel frequency-domain pathways using element-wise addition. A subsequent convolutional layer, followed by GN, is applied to stably normalize the fused features.

Finally, the features $\mathbf{f}_{\text{DFAI}}$ and $\mathbf{f}_{\text{fre-spa-g}}$ are concatenated to obtain features $\mathbf{F}_{\text{DCTSPE}}$, which is subsequently fed into a DBF module. Each branch of the DBF module consists of average pooling, convolution, GELU, and a sigmoid activation function. Thus, we have

$$\mathbf{f}_{\text{DFAI}} = \text{GN}\left(\text{Conv}_{3\times 3}\left(\mathbf{f}_{\text{DSI}} + \mathbf{f}_{\text{DCI}}\right)\right). \tag{28}$$

By filtering redundant features to capture stable high-frequency representations, the dual-path design greatly improves the model's capability to distinguish spectrally similar samples from different categories.

The spectral features $\mathbf{F}_{\text{spe}}$ are obtained by adding $\mathbf{f}_{\text{spe-local}}$ and $\mathbf{f}_{\text{fre-spe-g}}$, and the spatial features $\mathbf{F}_{\text{spa}}$ are formed by adding $\mathbf{f}_{\text{spa-local}}$ and $\mathbf{f}_{\text{fre-spa-g}}$. These two feature representations are subsequently fed into the DBF module for further aggregation as follows:

$$\mathbf{F}_{\text{spe}} = \mathbf{f}_{\text{spe-local}} + \mathbf{f}_{\text{fre-spe-g}} \tag{29}$$

$$\mathbf{F}_{\text{spa}} = \mathbf{f}_{\text{spa-local}} + \mathbf{f}_{\text{fre-spa-g}}. \tag{30}$$

The DBF module adopts a symmetric dual-branch architecture to process spectral and spatial features separately. Each branch is composed of average pooling and convolutional layers, along with GELU and a sigmoid activation function. The final output $\mathbf{F}_{\text{DBF}}$ of the module is obtained as follows:

$$\mathbf{F}'_{\text{spe}} = \sigma\left(\text{Conv}\left(\text{GELU}\left(\text{Conv}\left(\text{AvgPool}\left(\mathbf{F}_{\text{spe}}\right)\right)\right)\right)\right) \tag{31}$$

$$\mathbf{F}'_{\text{spa}} = \sigma\left(\text{Conv}\left(\text{GELU}\left(\text{Conv}\left(\text{AvgPool}\left(\mathbf{F}_{\text{spa}}\right)\right)\right)\right)\right) \tag{32}$$

$$\mathbf{F}_{\text{DBF}} = \left(\mathbf{F}'_{\text{spe}} * \mathbf{F}_{\text{spe}}\right) + \left(\mathbf{F}'_{\text{spa}} * \mathbf{F}_{\text{spa}}\right). \tag{33}$$

TABLE I
TRAINING, VALIDATION, AND TESTING SAMPLE SIZES OF THE HUST2013 DATASET

NO.	Category	Training	Validation	Testing
1	Healthy Grass	30	30	1191
2	Stressed Grass	30	30	1194
3	Synthetic Grass	30	30	637
4	Trees	30	30	1184
5	Soil	30	30	1182
6	Water	30	30	265
7	Residential	30	30	1208
8	Commercial	30	30	1184
9	Road	30	30	1192
10	Highway	30	30	1167
11	Railway	30	30	1175
12	Parking Lot1	30	30	1173
13	Parking Lot2	30	30	409
14	Tennis Court	30	30	368
15	Running Track	30	30	600
	Total	450	450	14129

TABLE II
TRAINING, VALIDATION, AND TESTING SAMPLE SIZES OF THE LK DATASET

NO.	Category	Training	Validation	Testing
1	Corn	30	30	34451
2	Cotton	20	20	8334
3	Sesame	20	20	2991
4	Broad-leaf soybean	30	30	63152
5	Narrow-leaf soybean	20	20	4111
6	Rice	20	20	11814
7	Water	30	30	66996
8	Roads and houses	25	25	7074
9	Mixed weed	25	25	5179
	Total	220	220	204102

Subsequently, the extracted feature map $\mathbf{F}_{\text{DBF}}$ is fed into an adaptive average pooling layer. The outputs are sequentially processed by a flatten layer, a ReLU-activated fully connected layer, a dropout layer, and a final linear layer to produce classification predictions.

IV. EXPERIMENTS

In this section, we provide a comprehensive evaluation of the proposed method in HSIC.

A. Dataset Description

To evaluate the effectiveness of the proposed model, we conducted experiments on three widely used HSI benchmark datasets: Houston 2013 (HUST2013), WHU-Hi-HongHu (HH), and WHU-Hi-LongKou (LK).

1) *HUST2013:* This dataset was acquired by the National Center for Airborne Laser Mapping (NCALM) in 2013 over the University of Houston campus and its neighboring urban area. It consists of 144 spectral bands with a spatial dimension of 349 × 1905 pixels and a spatial resolution of 2.5 m, encompassing 15 ground-truth classes, as detailed in Table I.
2) *LK:* This dataset was captured in July 2018 over an agricultural region in Longkou Town, Hubei Province,

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TABLE III
TRAINING, VALIDATION, AND TESTING SAMPLE SIZES OF THE HH DATASET

NO.	Category	Training	Validation	Testing
1	Red roof	44	44	13953
2	Road	44	44	3424
3	Bare soil	44	44	21733
4	Cotton	44	44	163197
5	Cotton firewood	44	44	6130
6	Rape	44	44	44469
7	Chinese cabbage	44	44	24015
8	Pakchoi	44	44	3966
9	Cabbage	44	44	10731
10	Tuber mustard	44	44	12306
11	Brassica parachinensis	44	44	10927
12	Brassica chinensis	44	44	8866
13	Small Brassica chinensis	44	44	22419
14	Lactuca sativa	44	44	7268
15	Celtuce	44	44	914
16	Film covered lettuce	44	44	7174
17	Romaine lettuce	44	44	2922
18	Carrot	44	44	3129
19	White radish	44	44	8624
20	Garlic sprout	44	44	3398
21	Broad bean	44	44	1240
22	Tree	44	44	3952
	Total	968	968	384757

China. It comprises 270 spectral bands with a spatial dimension of 550 × 400 pixels and a spatial resolution of approximately 0.463 m, containing nine ground-truth classes. The training, validation, and testing sample sizes of the LK dataset are shown in Table II.

3) *WHU-Hi-HH Dataset:* The HH dataset was collected in November 2017 over an agricultural area in Honghu City, Hubei Province, China. It contains 270 spectral bands with a spatial dimension of 940 × 475 pixels and an ultrahigh spatial resolution of 0.043 m, covering 22 ground-truth classes. Table III presents the detailed sample division of the HH dataset.

B. Experimental Setup and Parameter Configuration

All experiments were conducted on a computing platform equipped with an Intel Core i9-12900KF CPU, 24 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU. The software environment was configured using Python 3.9 and PyTorch 2.6.

In our experiments, the HSIs from the three datasets were partitioned into fixed-size patches: 11×11 for the HUST2013 dataset, 11×11 for the LK dataset, and 7×7 for the HH dataset. Here, the sampling rates for the training sets were empirically set to 3%, 0.1%, and 0.25% for the HUST2013, LK, and HH datasets, respectively. The training, validation, and testing sample sizes for each dataset are detailed in Tables I–III. For the proposed model, the learning rate was fixed at 0.005 across all datasets, and the batch size was 32 for HUST2013 and LK, and 64 for HH. These hyperparameter values were determined based on preliminary experimental tuning of the proposed model. To mitigate statistical variance introduced by random initialization, each independent experiment was repeated ten times, with each run trained for 200 training epochs. All final evaluation metrics were presented as mean ± standard deviation. To quantitatively evaluate the classification performance of the proposed method, we employed overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (Kappa). In general, higher values of these metrics indicate better classification performance.

C. Comparative Experiments

To evaluate the superiority of the proposed method, ten state-of-the-art HSIC methods were employed for comparison: HybridFormer [41], SQSFormer [42], DCTN [28], MSFICNet [34], MHCFormer [32], DSFormer [25], MSSTnorm [43], EATN [27], HyperRope [26], and DEMUNet [30]. To ensure a fair evaluation, the hyperparameters of all baseline models were set to the same configuration, thereby eliminating experimental bias caused by different parameter settings.

1) Quantitative Analysis: Tables IV–VI comprehensively present the evaluation results (i.e., OA, AA, and Kappa) of various comparative algorithms on the HUST2013, LK, and HH datasets. All results are presented as the mean ± standard deviation across ten independent experiments, and the best-performing results are highlighted in boldface. The experimental data indicate that the proposed FBCANet method demonstrates comprehensively superior classification performance across all three datasets. Specifically, the proposed model achieves impressive OA values of 96.07%, 99.01%, and 92.04% on the HUST2013, LK, and HH datasets, respectively. Compared with ten state-of-the-art methods, our proposed FBCANet method achieves remarkable gains in classification accuracy while yielding the lowest standard deviation. This superior performance demonstrates its strong feature discrimination and outstanding stability.

We can see from Table IV that EATN achieves the best performance among all comparison methods. This advantage stems from the self-similarity feature enhancement mechanism, which significantly enhances the spatial representation capability of original feature patches. Nevertheless, the proposed FBCANet method outperforms its competitors on the HUST2013 dataset with a remarkably small standard deviation. This is mainly attributed to the effective modeling capability of the FDSSGA module. Different from methods that only adopt spectral and spatial enhancement, the FDSSGA module explicitly decouples complex frequency representations to reduce redundant interference. Meanwhile, a bias-driven calibration strategy is embedded to dynamically refine global contextual features. This design effectively mitigates spectral confusion arising from similar spectral signatures across different classes, which is particularly prevalent in urban scenes.

The classification results on the LK dataset are presented in Table V, where the AA of the proposed FBCANet method outperforms the second-best method by at least 0.81%. Notably, the LK dataset encompasses fine-grained agricultural scenes, where crops such as rice, corn, and soybeans exhibit highly similar spectral characteristics during their growth periods. Furthermore, intersecting linear field ridges result in blurred boundaries, thereby rendering the overall classification task extremely challenging. Even under these conditions, our

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TABLE IV
EXPERIMENTAL RESULTS ON THE HUST2013 DATASET

No.	HybridFormer	SQSFormer	DCTN	MSFICNet	MHCFormer	DSFormer	MSSTnorm	EATN	HyperRope	DEMUNet	Ours
1	**95.11±0.58**	93.68±0.36	94.69±0.65	95.04±0.83	94.51±1.16	94.52±0.30	94.67±2.22	95.08±0.51	93.79±0.24	94.18±0.67	94.01±0.49
2	95.85±0.93	92.34±1.55	94.50±1.59	95.43±0.94	93.45±2.25	95.59±1.58	95.66±1.05	**98.56±0.68**	93.63±1.29	96.37±1.04	96.39±1.78
3	**100.00±0.00**	99.97±0.07	99.92±0.15	**100.00±0.00**	99.84±0.18	**100.00±0.00**	98.86±0.91	**100.00±0.00**	99.98±0.05	**100.00±0.00**	**100.00±0.00**
4	**99.33±0.30**	95.55±1.31	99.13±0.31	95.47±1.50	98.43±0.44	95.97±0.70	95.82±2.89	97.17±0.78	97.51±0.59	96.00±0.84	97.27±0.61
5	95.94±1.59	98.23±0.82	95.87±0.77	96.10±1.05	97.68±1.24	96.56±0.51	93.18±1.22	97.84±0.33	**98.92±0.77**	95.26±1.70	97.84±1.15
6	91.32±3.60	95.25±1.69	88.29±2.37	87.25±3.44	82.54±6.59	82.68±3.22	87.97±7.74	92.82±5.26	89.03±3.32	93.67±4.56	**96.69±1.66**
7	95.08±0.80	93.11±0.95	95.13±0.98	94.21±1.57	95.42±1.19	89.81±0.64	90.11±1.69	95.74±0.82	94.65±1.43	**96.78±0.73**	96.19±1.26
8	95.33±0.98	94.89±1.33	96.61±1.06	94.76±2.24	92.70±2.14	**98.63±0.81**	84.95±2.63	96.13±0.80	96.12±1.94	96.12±1.12	98.52±0.69
9	89.16±1.50	92.12±1.10	92.50±1.40	94.11±1.05	97.26±0.66	94.91±0.81	90.31±2.50	93.46±1.02	91.59±1.77	95.59±1.16	**97.03±1.29**
10	83.54±1.31	88.36±1.31	87.15±1.58	92.80±1.11	**95.08±1.29**	95.47±0.61	89.62±1.99	92.17±1.38	88.92±1.98	92.39±1.82	94.29±1.55
11	86.61±0.99	92.36±2.47	91.62±2.45	92.79±1.26	95.31±1.25	94.71±0.76	86.03±2.26	**95.44±1.04**	93.86±1.60	93.44±2.15	94.91±1.31
12	83.87±0.91	87.44±2.84	80.53±2.66	87.04±3.55	80.85±1.31	92.57±0.78	83.28±2.41	93.44±1.24	88.90±1.69	85.68±1.62	**95.74±1.45**
13	94.15±2.40	93.94±2.17	86.66±2.03	83.92±3.35	89.45±2.33	76.08±1.63	71.18±7.25	91.27±2.99	**94.73±3.56**	92.78±4.03	91.37±2.02
14	88.59±5.54	97.35±1.68	94.79±3.96	97.69±1.88	86.14±5.08	95.04±3.59	81.92±4.00	98.51±2.12	**98.86±1.39**	95.86±2.94	95.82±2.21
15	96.06±1.19	95.76±1.16	94.77±0.72	94.66±1.28	**97.27±1.71**	89.66±1.44	96.81±1.44	95.37±0.87	95.23±0.92	96.88±1.95	92.73±1.31
OA	92.28±0.39	93.39±0.50	92.64±0.37	93.65±0.41	93.61±0.51	93.83±0.18	90.00±0.61	95.57±0.13	94.10±0.47	94.42±0.37	**96.07±0.12**
AA	92.66±0.44	94.02±0.53	92.81±0.38	93.42±0.47	93.06±0.64	92.81±0.28	89.36±0.78	95.53±0.26	94.38±0.59	94.73±0.51	**95.92±0.18**
Kappa	91.64±0.42	92.85±0.54	92.04±0.40	93.13±0.45	93.09±0.55	93.33±0.20	89.19±0.65	95.21±0.14	93.61±0.51	93.97±0.40	**95.75±0.20**

TABLE V
EXPERIMENTAL RESULTS ON THE LK DATASET

No.	HybridFormer	SQSFormer	DCTN	MSFICNet	MHCFormer	DSFormer	MSSTnorm	EATN	HyperRope	DEMUNet	Ours
1	99.43±0.06	99.12±0.25	99.36±0.11	99.23±0.21	98.90±0.75	96.53±0.41	94.03±2.68	**99.51±0.07**	98.79±0.28	99.45±0.15	99.16±0.15
2	96.60±0.72	90.73±3.06	94.33±1.04	98.49±0.75	96.38±0.83	84.62±0.62	73.87±9.09	93.93±1.72	94.70±1.82	99.02±0.94	**99.57±0.17**
3	94.27±2.00	84.74±3.06	94.09±2.31	85.35±4.00	94.01±1.82	80.74±1.79	27.62±13.37	92.27±1.85	87.36±3.75	87.67±5.64	**97.12±1.07**
4	**99.85±0.04**	99.44±0.07	99.36±0.09	99.16±0.21	99.73±0.06	98.46±0.04	96.81±1.30	99.67±0.10	99.55±0.10	99.58±0.17	99.73±0.09
5	77.03±4.08	77.45±4.01	65.84±4.10	78.37±3.74	82.40±4.61	72.05±1.41	55.20±3.97	**90.34±1.39**	78.84±5.87	79.55±4.60	86.80±5.37
6	97.74±1.01	96.95±1.35	97.96±0.61	99.42±0.48	98.82±0.58	95.03±0.65	93.20±2.32	99.55±0.17	97.09±0.80	99.04±0.46	**99.70±0.13**
7	99.95±0.01	99.96±0.01	**99.96±0.00**	99.96±0.01	99.96±0.02	99.31±0.13	99.64±0.19	99.96±0.01	99.96±0.03	99.80±0.24	**99.96±0.00**
8	93.72±0.60	94.16±0.97	94.22±1.28	91.91±1.23	93.00±1.23	86.10±0.78	76.95±4.39	96.15±0.47	94.83±1.22	93.04±1.06	**97.23±0.62**
9	**96.80±0.49**	91.66±2.76	95.13±0.60	88.79±2.09	88.64±2.07	83.40±1.06	75.26±4.57	90.69±1.25	93.53±2.20	92.89±2.48	90.08±1.84
OA	98.58±0.20	97.83±0.29	97.91±0.24	98.13±0.17	98.42±0.16	95.81±0.09	90.20±1.36	98.80±0.08	98.16±0.24	98.45±0.27	**99.01±0.18**
AA	95.04±0.59	92.69±0.87	93.36±0.76	93.41±0.69	94.65±0.51	88.47±0.22	76.95±1.14	95.78±0.29	93.85±0.92	94.45±0.90	**96.59±0.69**
Kappa	98.14±0.26	97.16±0.38	97.26±0.31	97.55±0.22	97.93±0.21	94.52±0.12	87.36±1.70	98.42±0.10	97.58±0.32	97.97±0.35	**98.70±0.23**

TABLE VI
EXPERIMENTAL RESULTS ON THE HH DATASET

No.	HybridFormer	SQSFormer	DCTN	MSFICNet	MHCFormer	DSFormer	MSSTnorm	EATN	HyperRope	DEMUNet	Ours
1	98.51±0.28	98.34±0.38	98.11±0.23	98.91±0.44	96.18±0.55	97.27±0.21	79.07±9.97	**98.99±0.22**	98.23±0.36	96.30±1.18	97.77±0.37
2	81.71±1.62	80.36±2.49	74.67±3.58	82.51±1.90	77.06±2.95	75.31±3.20	53.72±3.87	**83.12±1.48**	80.96±2.04	80.52±2.39	77.43±1.92
3	97.20±0.49	96.57±0.71	94.95±0.81	97.58±0.59	96.12±0.77	93.46±0.77	88.54±2.24	96.19±0.74	97.06±0.57	98.18±0.69	**98.22±0.44**
4	99.75±0.04	99.68±0.09	99.09±0.13	99.70±0.06	99.46±0.06	99.37±0.14	98.56±0.31	99.74±0.03	99.65±0.07	99.65±0.07	**99.79±0.03**
5	55.88±3.58	43.13±4.32	49.82±6.21	53.86±4.32	39.36±3.23	30.70±2.59	16.00±5.99	36.97±1.77	55.28±8.67	51.16±3.10	**59.78±3.72**
6	95.78±0.24	95.72±0.34	92.62±0.92	96.33±0.95	94.11±0.74	88.99±1.95	88.75±2.92	96.07±0.45	96.43±0.35	96.07±0.33	**97.22±0.18**
7	90.38±1.12	89.47±1.40	89.36±1.23	89.02±1.13	89.21±0.93	84.38±3.20	69.90±4.13	**92.79±1.23**	88.29±1.52	90.48±1.85	91.06±0.72
8	50.64±2.16	50.25±2.68	31.93±1.64	45.26±3.04	41.48±3.94	26.89±2.06	14.90±2.74	58.69±3.90	46.93±2.31	43.07±3.61	**59.60±3.24**
9	99.15±0.24	98.79±0.52	98.24±0.64	99.01±0.57	95.01±1.16	98.38±0.39	87.39±2.50	99.20±0.22	99.07±0.34	99.44±0.31	**99.68±0.16**
10	73.91±2.42	75.20±2.90	77.22±2.91	79.39±1.60	74.07±2.43	60.84±7.35	67.34±2.01	81.28±2.47	75.01±2.75	**84.36±2.05**	80.47±3.21
11	73.47±2.37	69.96±2.98	64.79±2.91	63.87±2.79	67.21±2.20	62.98±6.29	39.77±4.24	69.67±2.74	68.02±2.55	70.20±1.81	**79.60±2.58**
12	65.59±1.63	56.20±2.16	57.61±3.05	60.94±2.78	62.33±2.06	43.34±4.25	49.26±4.42	57.10±2.01	57.39±2.31	59.79±2.00	**67.30±2.69**
13	82.69±1.02	79.80±0.83	73.95±1.09	81.78±1.94	78.50±2.34	72.68±2.26	70.43±4.11	**87.34±0.95**	78.99±1.50	84.45±2.04	87.28±1.33
14	92.04±2.00	90.63±2.51	84.56±2.57	91.76±1.34	76.91±2.98	71.32±9.27	53.59±9.09	93.93±2.31	90.96±2.61	94.22±1.64	**96.68±1.01**
15	74.34±3.31	57.75±6.68	50.98±4.74	69.66±2.85	37.78±7.18	40.70±7.12	17.38±4.48	77.24±8.77	68.12±5.35	60.77±12.65	**85.72±7.00**
16	97.79±0.73	98.20±0.55	93.00±1.55	95.09±0.53	91.83±1.67	84.01±3.69	77.38±5.42	**98.91±0.25**	96.87±1.48	97.57±1.20	98.70±0.32
17	88.49±1.34	84.56±2.27	63.27±3.70	87.61±2.05	79.49±2.59	59.04±7.62	40.60±5.96	86.29±2.85	86.72±2.26	89.76±1.53	**94.13±1.82**
18	81.09±2.83	78.90±1.94	63.73±2.56	72.56±4.54	53.51±3.35	67.14±3.13	34.32±3.61	**86.93±2.22**	78.94±4.98	77.58±4.36	84.25±2.87
19	81.20±1.47	79.35±1.81	77.48±2.65	77.71±2.76	77.98±2.90	69.69±2.49	58.00±6.09	80.10±1.57	78.50±3.83	78.44±2.79	**84.62±1.13**
20	74.92±1.42	82.54±2.77	63.97±1.88	79.07±1.51	69.72±2.45	64.00±2.60	42.85±6.54	74.02±4.48	79.61±5.30	77.68±3.62	**86.87±2.09**
21	37.52±3.95	29.60±2.20	15.51±2.70	40.33±5.46	28.14±2.55	17.41±3.40	15.77±2.60	**56.26±4.99**	25.84±2.36	38.51±4.94	45.27±5.78
22	47.31±3.48	48.18±4.51	39.70±1.88	40.83±4.98	37.92±2.96	48.72±3.62	13.17±3.62	47.17±2.84	50.92±2.94	**54.39±5.65**	52.65±3.71
OA	90.11±0.20	88.32±0.46	85.65±0.59	88.90±0.58	85.89±0.70	82.03±1.65	69.15±2.28	89.07±0.35	88.82±0.64	89.81±0.40	**92.04±0.24**
AA	79.06±0.24	76.51±0.61	70.66±0.71	77.40±0.71	71.06±1.05	66.21±2.80	53.49±1.91	79.91±0.77	77.17±0.64	78.30±0.84	**82.91±0.56**
Kappa	87.64±0.24	85.45±0.55	82.08±0.71	86.16±0.69	82.51±0.83	77.72±2.04	63.30±2.47	86.47±0.41	86.03±0.75	87.28±0.48	**90.03±0.30**

FBCANet method leverages the DFAI module to precisely capture and compensate for the high-frequency edges and fine-grained textures of small-scale crops. This effectively amplifies spectral differences among highly similar crops, greatly mitigating classification boundary aliasing arising from categories with similar spectra.

Table VI shows the performance of various HSIC methods on the HH dataset, which possesses the most complex

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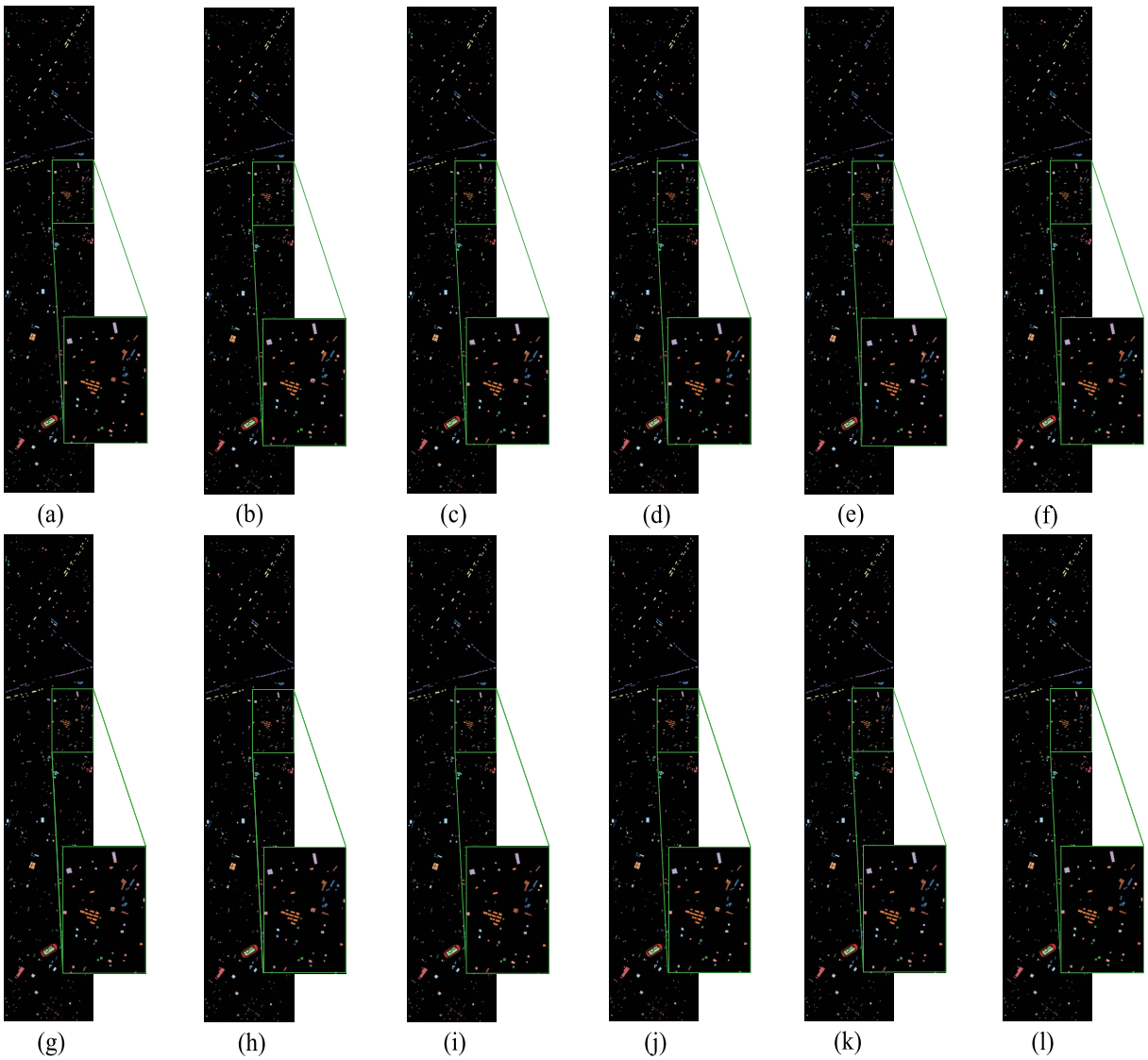


Fig. 4. Classification maps of different methods on the HUST2013 dataset. (a) GT. (b) HybridFormer. (c) SQSFormer. (d) DCTN. (e) MSFICNet. (f) MHCFormer. (g) DSFormer. (h) MSSTnorm. (i) EATN. (j) HyperRope. (k) DEMUNet. (l) Ours.

category distribution. It can be seen that HybridFormer achieves the second-best result among all comparison methods on this dataset. The main reason is that it adopts a hybrid transformer, which integrates multigranularity tokens and a spatial–spectral attention (SSA) mechanism to focus on more differentiated channels and spatial locations. In contrast, the OA of our FBCANet method is further improved by 1.93%. The superior performance is attributed to the complementary design and seamless integration of the LSFE, FDSSGA, and DCTLSFE modules. Specifically, the LSFE module efficiently captures local spectral features through lightweight convolutions. The FDSSGA module utilizes FFT to adaptively recalibrate frequency components, suppress redundant information, and integrate global context, thereby improving feature discriminability. Meanwhile, the DCTLSFE module explicitly exploits high-frequency information to capture fine-grained spatial textures and different category boundaries. Together, they enable a robust and discriminative representation across spatial–spectral and local–global domains, thereby leading to consistent improvements in classification accuracy.

2) Qualitative Analysis: Figs. 4–6 present the classification maps produced by different HSIC methods on three HSI datasets. The proposed FBCANet method demonstrates superior performance compared to other models in delineating complex land-cover boundaries and classifying small-scale regions. Its competitors exhibit severe misclassification near the boundaries of spectrally similar classes. By leveraging the explicit decoupling and refinement of spatial–spectral information facilitated by the frequency-domain mechanisms, our FBCANet method effectively suppresses redundant interference. As a result, the classification maps exhibit smoother boundaries and better consistency with the ground truth, which further demonstrates the superiority of our proposed method.

D. Hyperparameter Analysis

This section primarily investigates the influence of key hyperparameters, including patch size, batch size, and learn-

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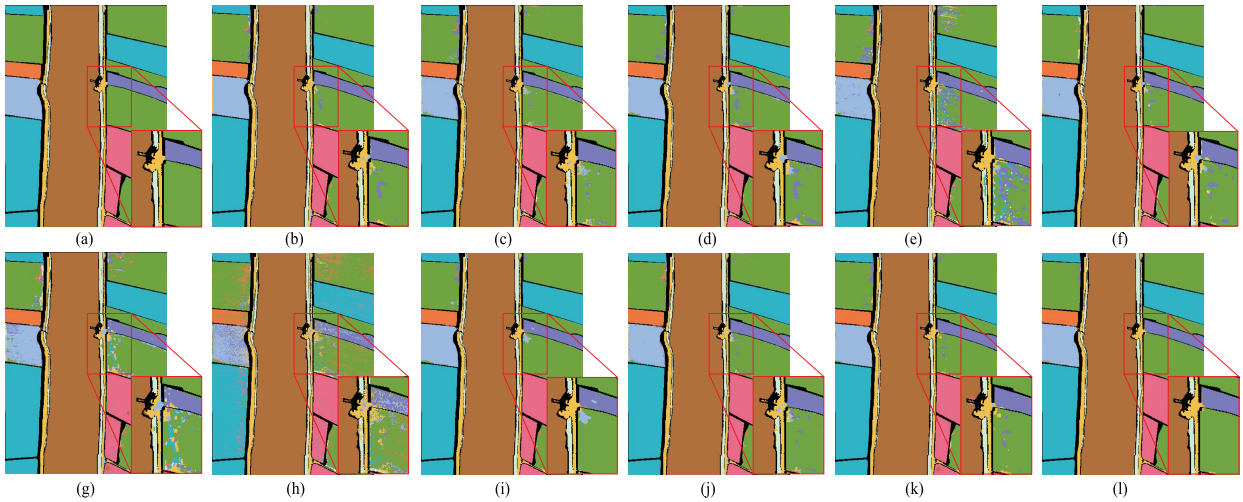


Fig. 5. Classification maps of different methods on the LK dataset. (a) GT. (b) HybridFormer. (c) SQSFormer. (d) DCTN. (e) MSFICNet. (f) MHCFormer. (g) DSFormer. (h) MSSTnorm. (i) EATN. (j) HyperRope. (k) DEMUNet. (l) Ours.

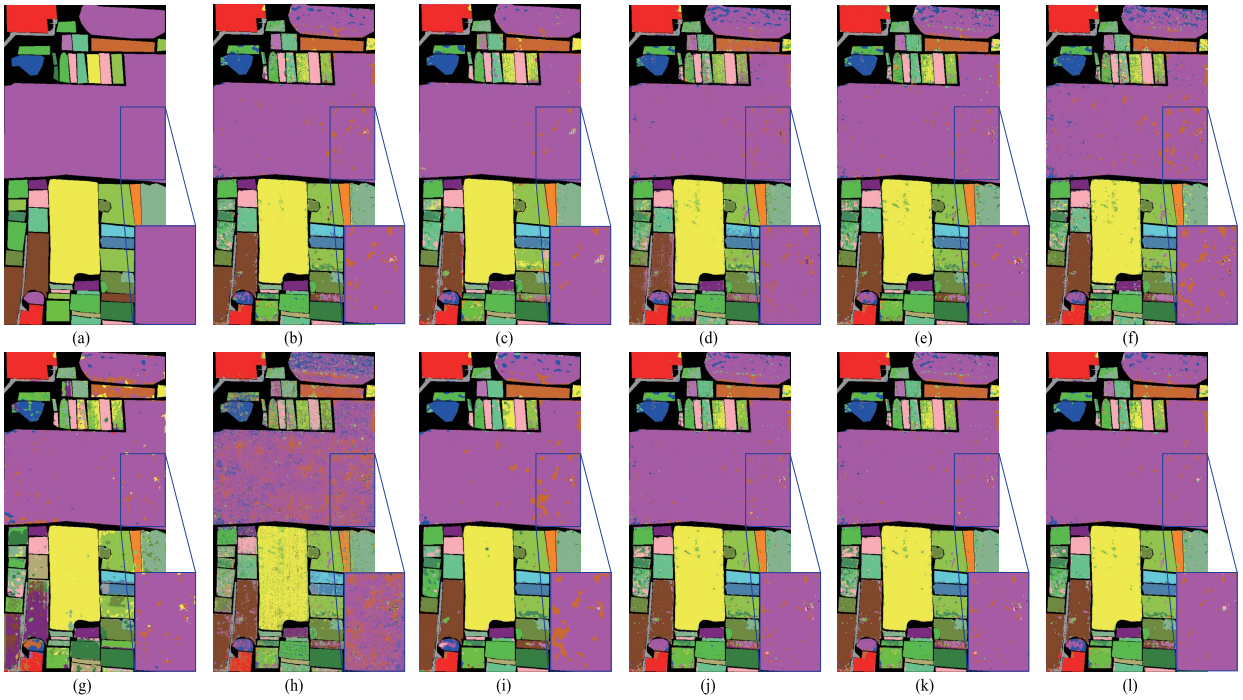


Fig. 6. Classification maps of different methods on the HH dataset. (a) GT. (b) HybridFormer. (c) SQSFormer. (d) DCTN. (e) MSFICNet. (f) MHCFormer. (g) DSFormer. (h) MSSTnorm. (i) EATN. (j) HyperRope. (k) DEMUNet. (l) Ours.

ing rate, on classification performance. The implementation follows a controlled variable strategy: one hyperparameter was varied at a time, while all other settings were fixed at their optimal values. Table VII compares the computational complexity of different models on the HUST2013 dataset. The detailed experimental results for each hyperparameter are presented as follows.

1) *Patch Size:* In this study, we deeply explored the influence of patch size on the model's classification performance. Theoretically, a larger patch size facilitates the extraction of richer spatial context information. However, an excessively large receptive field may introduce redundant information interference, thereby degrading classification accuracy. With all other hyperparameters fixed, we performed comparative experiments with patch sizes of {5, 7, 9, 11, 13}. As illustrated in Fig. 7, the classification accuracy across the three datasets rises first and then decreases. These results demonstrate that an optimal receptive field range exists that adapts to the unique characteristics of different HSI datasets. Specifically, the optimal patch sizes for the three datasets are empirically determined as 11, 11, and 7, respectively.
2) *Batch Size:* During model training, the batch size directly influences gradient estimation and the underlying training dynamics. Gradient variance arising from

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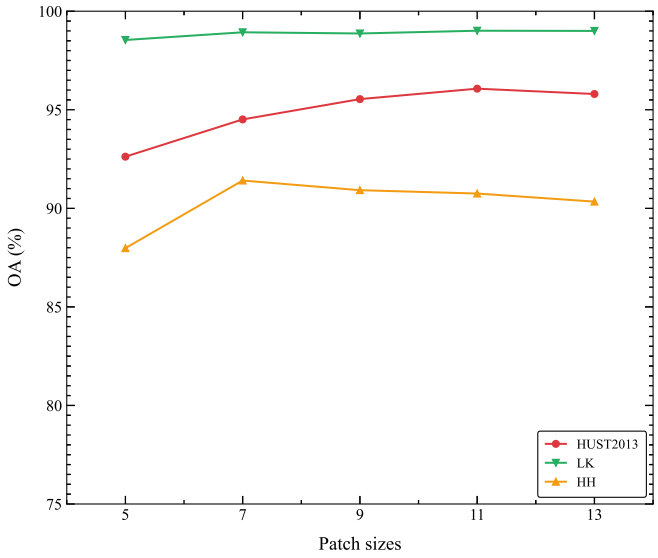


Fig. 7. OA comparison at different patch sizes.

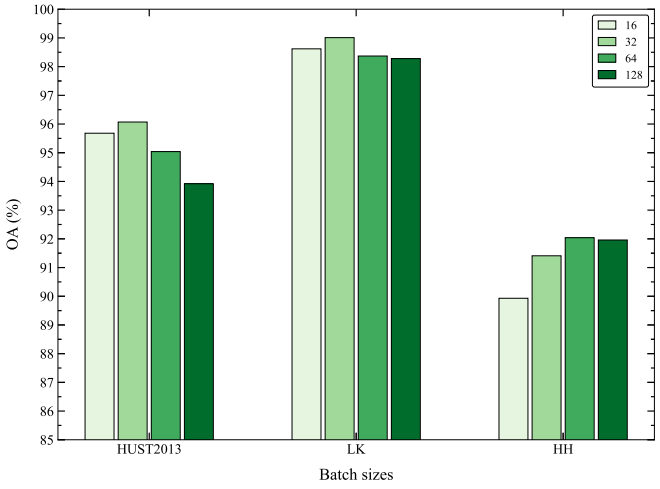


Fig. 8. OA comparison at different batch sizes.

small-batch training introduces an implicit regularization effect, facilitating convergence toward flatter minima and boosting model generalization. Although large batch sizes can improve numerical stability and convergence efficiency, they often compromise the model's generalization performance. Therefore, we systematically evaluated the model's sensitivity to batch sizes ranging from 16 to 128. It can be observed from the experimental results in Fig. 8 that setting the batch size to 32 yields the best balance between classification accuracy and training stability for the HUST2013 and LK datasets. Meanwhile, optimal performance on the HH dataset is attained with a batch size of 64.

3) *Learning Rate:* To determine the optimal learning rate, we evaluated four candidate values: 0.001, 0.005, 0.01, and 0.05. The experimental results (Fig. 9) show that classification accuracy presents an upward-then-downward trend with changing learning rates, peaking at 0.005. This phenomenon verifies the classic trade-off

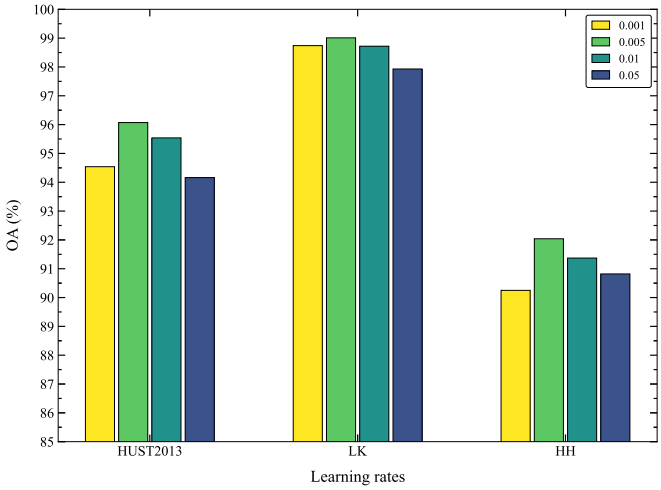


Fig. 9. OA comparison at different learning rates.

in model optimization. An overly small step size slows convergence and restricts generalization capability, while an excessively large step size impairs gradient stability. Here, we empirically set the learning rate to 0.005 to achieve rapid convergence and sustain stable training for the proposed classification model.

E. Ablation Study

To verify the specific contributions of each core module within the proposed model to overall classification performance, comprehensive ablation studies were conducted on the HUST2013 dataset. The quantitative results of these experiments are shown in Table VIII, where "×" indicates that the module is disabled, "✓" indicates it is activated, and "–" denotes that the module is replaced with a functionally equivalent alternative.

We designed a step-wise ablation study to evaluate the impact of each component. A minimalist framework containing only the DBF fusion module serves as the baseline, achieving an OA of 86.86% (Case 1). Subsequently, the GFEF, BCGA, DFAI, and SpecBlock modules are integrated individually into the baseline (Cases 2–5) to quantify the performance improvement of each standalone component. Empirical results demonstrate that the BCGA, DFAI, and SpecBlock modules independently increase the OA to approximately 93.86%, fully validating their efficiency in spatial–spectral feature extraction. Furthermore, a direct comparison between Case 2 and Case 6 explicitly confirms that substituting the standard multihead attention mechanism with the proposed BCGA module yields a notable performance gain. This improvement is primarily attributed to its structured additive operations, which preserve a global modeling receptive field and effectively extract discriminative features of HSIs.

To evaluate the interaction between modules, Case 6 cascades the GFEF and BCGA modules, yielding substantial improvements in OA, AA, and the Kappa coefficient of 6.30%, 5.33%, and 6.81%, respectively. This indicates that after GFEF performs preliminary feature screening and denoising via frequency analysis, it integrates with the BCGA module to form

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TABLE VII
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT MODELS ON THE HUST2013 DATASET

Indicators	HybridFormer	SQSFormer	DCTN	MSFICNet	MHCFormer	DSFormer	MSSTnorm	EATN	HyperRope	DEMUNet	Our
FLOPs (G)	2.95	0.84	**29.20**	1.10	10.83	0.66	1.06	3.70	0.90	23.18	5.06
Params(M)	2.76	0.23	**68.30**	0.37	2.78	2.79	0.48	0.96	0.24	28.58	1.44
Memory(MB)	56.00	116.63	388.27	104.50	129.00	47.60	56.93	80.32	284.18	**528.92**	59.20
Infer time (S)	3.09	0.59	**20.44**	7.22	2.37	3.12	3.01	2.02	1.10	10.05	6.77
Train time (S)	23.86	4.69	**76.03**	40.33	19.38	14.17	7.56	19.64	5.30	41.33	36.57
OA (%)	92.28	93.39	92.64	93.65	93.61	93.83	90.07	95.51	94.10	94.42	**96.07**

TABLE VIII
ABLATION EXPERIMENTAL RESULTS ON THE HUST2013 DATASET

Case	FDSSGA		DCTLSFE	LSFE	Fusion	Evaluation metrics		
	GFEF	BCGA	DFAI	SpecBlock	DBF	OA	AA	Kappa
1	×	×	×	×	✓	86.86	87.42	85.79
2	✓	–	×	×	✓	87.64	88.29	86.64
3	×	✓	×	×	✓	93.56	94.10	93.03
4	×	×	✓	×	✓	93.66	93.17	93.14
5	×	×	×	✓	✓	94.36	94.01	93.90
6	✓	✓	×	×	✓	93.94	93.62	93.45
7	✓	✓	✓	×	✓	95.19	95.05	94.79
8	✓	✓	✓	✓	–	95.72	95.61	95.37
9	–	✓	✓	✓	✓	94.79	94.54	94.36
10	–	✓	✓	✓	✓	95.88	95.57	95.54
11	✓	–	✓	✓	✓	95.73	95.58	95.38
12	✓	✓	✓	✓	✓	**96.07**	**95.92**	**95.75**

a highly efficient frequency-domain structured spatial–spectral global aggregation mechanism, thereby significantly optimizing feature representations. Following this, Case 7 introduces the DFAI module into the pipeline. By leveraging the sparsity of the transform domain to capture high-frequency edges and fine-grained textures, the DFAI module achieves more effective extraction of intricate local spatial details, yielding additional gains of 1.25%, 1.43%, and 1.34% across the three metrics. To further analyze the internal mechanisms of the GFEF module, it can be observed that removing the dual attention mechanism in Case 9 yields performance degradations of 1.28%, 1.38%, and 1.39%, respectively. This performance decline demonstrates that dual attention effectively enhances the global representation capability. Similarly, Case 10 replaces the separate processing of real and imaginary components in the GFEF module with joint complex-valued processing. This modification causes decreases of 0.19%, 0.35%, and 0.21% in the three evaluation metrics. For the BCGA module, Case 11 replaces the additive bias fusion strategy with multiplicative fusion for generating B_{bias}, leading to degraded classification performance. This result suggests that additive bias fusion can better preserve complementary information by adaptively adjusting feature responses, whereas multiplicative fusion may introduce excessive feature attenuation and weaken discriminative representations. Finally, Case 8 evaluates the effectiveness of the DBF module by replacing it with a naive feature-addition strategy. Compared with this simplified fusion scheme, the complete model (Case 12) achieves optimal classification performance: 96.07% OA, 95.92% AA, and 95.75% (Kappa). Overall, these modules work synergistically to achieve optimal classification results.

F. Computational Complexity Analysis

To evaluate the computational efficiency of the proposed model, we compare its resource consumption with that of ten HSIC methods. As reported in Table VII, experiments conducted on the HUST2013 dataset assess floating-point operations (FLOPs), parameter count (Params), memory usage (Memory), and OA. Specifically, the proposed model requires 5.06 G FLOPs and 1.44 M parameters for training. It should be clarified that this parameter count comprehensively includes all trainable weights across the entire backbone network and the final classification layer. Although it incurs a moderately higher computational cost than ultra-lightweight networks such as SQSFormer, this trade-off is justified by its superior classification performance. Compared with computationally intensive architectures, such as DCTN and DEMUNet, our method reduces computational overhead by an order of magnitude. Notably, its memory footprint of 59.20 MB further demonstrates its practicality for real-world deployment. Compared with computationally intensive models such as DCTN and DEMUNet, the proposed framework maintains superior classification performance while achieving reasonable training and inference efficiency. Overall, the proposed model achieves a favorable balance between computational efficiency and classification performance.

V. CONCLUSION

In this article, we propose a novel collaborative three-module HSIC framework, called FBCANet for HSIC. Specifically, the LSFE module is designed to capture continuous local spectral features through lightweight convolutions. The FDSSGA module is designed to model global dependencies and adaptively filter redundant spectral signatures induced by various interference sources. Within this module, the GFEF module explicitly decouples complex frequency representations via the Fourier Transform. Concurrently, the BCGA module generates a bias compensation component from local spatial–spectral representations and then injects it additively into the global interaction pathway. This design enables dynamic refinement of backbone features, thereby significantly enhancing interclass discriminability. In addition, the DCTLSFE module explicitly extracts high-frequency components across both spatial and channel dimensions. These components are subsequently refined using a spatial gating network and adaptive channel pooling to explore fine-grained

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textures and complex land-cover boundaries. Extensive experiments on three benchmark datasets demonstrate that the proposed FBCANet method consistently achieves state-of-the-art classification performance in practice HSIC tasks.

Although our FBCANet method achieves exceptional performance with sufficient training data, pixel-level annotation for HSIs is notoriously labor-intensive. Therefore, future work will extend this frequency-aware architecture to few-shot learning frameworks, aiming to achieve high classification accuracy with extremely limited annotated samples.

REFERENCES

[1] M. Zhu et al., "Application of hyperspectral technology in detection of agricultural products and food: A review," *Food Sci. Nutrition*, vol. 8, no. 10, pp. 5206–5214, Oct. 2020.

[2] W. Zhang and S. Liu, "Applications of the small satellite constellation for environment and disaster monitoring and forecasting," *Int. J. Disaster Risk Sci.*, vol. 1, no. 2, pp. 9–16, 2010.

[3] S. Peyghambari and Y. Zhang, "Hyperspectral remote sensing in lithological mapping, mineral exploration, and environmental geology: An updated review," *J. Appl. Remote Sens.*, vol. 15, no. 3, pp. 031501–031526, Jul. 2021.

[4] H. Yan, J. He, X. Zhang, and Z. Li, "Semantic anchor enhanced prototype learning for few-shot hyperspectral image classification," *Knowl.-Based Syst.*, vol. 341, pp. 115822–115839, May 2026.

[5] Y. Zhang, J. He, and Y. Li, "Cross-domain few-shot hyperspectral classification via orthogonal feature disentanglement," *Eng. Appl. Artif. Intell.*, vol. 169, pp. 114071–114087, Apr. 2026.

[6] Z. Zhong, C. Shi, W. Sun, K. Wang, and Y. Zhang, "Adaptive contourlet-guided fuzzy fusion network for joint classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5510412.

[7] C. Shi, Y. Wang, W. Sun, C. Fu, and Z. Jin, "HSMMamba: Hierarchical superpixel MoE with mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 5519414–5519428, 2026.

[8] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[9] L. Ma, M. M. Crawford, and J. Tian, "Local manifold learning-based k-nearest-neighbor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4099–4109, Nov. 2010.

[10] C. Cariou and K. Chehdi, "Unsupervised nearest neighbors clustering with application to hyperspectral images," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1105–1116, Sep. 2015.

[11] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, no. 1, pp. 1–12, 2015.

[12] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[13] A. Ben Hamida, A. Benoit, P. Lambert, and C. Ben Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[14] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[15] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[16] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[17] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415.

[18] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[19] R. Xu, X.-M. Dong, W. Li, J. Peng, W. Sun, and Y. Xu, "DBCTNet: Double branch convolution-transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509915.

[20] Z. Shu, Z. Liu, Z. Yu, and X.-J. Wu, "Dual feature aggregation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5502916.

[21] Z. Zhao, D. Hu, H. Wang, and X. Yu, "Convolutional transformer network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[22] B. Zhang, X. Dong, Z. Ye, and Q. Qu, "A wavelet frequency domain enhanced network for hyperspectral image classification," in *Proc. 8th Int. Conf. Comput. Inf. Sci. Appl. Technol. (CISAT)*, Jul. 2025, pp. 372–377.

[23] P. Liu, X. Fu, N. Xiao, R. He, S. Zhang, and S. Jia, "Frequency-enhanced spatial–spectral network for hyperspectral and multispectral image fusion," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Aug. 2025, pp. 8830–8834.

[24] J. Zhang, C. Zhang, S. Liu, Z. Shi, and B. Pan, "Three-dimensional frequency-domain transform network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5409813.

[25] Y. Xu, D. Wang, L. Zhang, and L. Zhang, "Dual selective fusion transformer network for hyperspectral image classification," *Neural Netw.*, vol. 187, pp. 107311–107324, Jul. 2025.

[26] Z. Khan, N. Dey, K. Kathiravan, S.-C. Yoon, and S. M. Bhandarkar, "Spatial–spectral transformer with patch-local mixed-axis 2-D rotary position embedding for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 1910–1925, 2026.

[27] Y. Wang, Z. Shu, and Z. Yu, "Efficient attention transformer network with self-similarity feature enhancement for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 11469–11486, 2025.

[28] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "DCTN: Dual-branch convolutional transformer network with efficient interactive self-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508616.

[29] X. Lu, M. Yang, Y. Chen, S. Xiong, and X. Lu, "Multibranch fusion-based feature enhance for remote-sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4702017.

[30] X. Cui and L. Zhang, "DEMUNet: Dual encoder mamba U-Net for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 99–113, 2026.

[31] Z. Shu, K. Zeng, J. Zhou, Y. Wang, M.-Y. Tai, and Z. Yu, "Spectral–spatial synergy guided attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5523316.

[32] H. Shi, Y. Zhang, G. Cao, and D. Yang, "MHCFormer: Multiscale hierarchical conv-aided Fourierformer for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.

[33] S. Fan, Q. Liu, W. Li, and H. Bai, "A frequency and topology interaction network for hyperspectral image classification," *Eng. Appl. Artif. Intell.*, vol. 133, pp. 108234–108248, Jul. 2024.

[34] Q. Liu, S. Fan, W. Li, S. Wei, and Y. Jiang, "A collaborative network via multi-head sparse and high-low frequency interaction for hyperspectral image classification," *Neurocomputing*, vol. 625, pp. 129541–129556, Apr. 2025.

[35] X. Wan, F. Chen, D. Mo, H. Liu, Z. Li, and K. Hu, "FS-CGNet: Frequency spectral–channel fusion and cross-scale global aggregation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5526222.

[36] Z. Chen, B. Tu, B. Liu, J. Li, and A. Plaza, "AF^{2}DN: Attention-guided frequency feature decomposition network for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5530716.

[37] L. Hu, J. Yuan, B. Cheng, and Q. Xu, "CSFPR-RTDETR: Real-time small object detection network for UAV images based on cross-spatial-frequency domain and position relation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5638219.

[38] T. Zhang et al., "CAS-ViT: Convolutional additive self-attention vision transformers for efficient mobile applications," *IEEE Trans. Image Process.*, vol. 35, pp. 1899–1909, 2026.

[39] A. Shaker, M. Maaz, H. Rasheed, S. Khan, M.-H. Yang, and F. S. Khan, "SwiftFormer: Efficient additive attention for transformer-based real-time mobile vision applications," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17379–17390.

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[40] Z. Shi et al., "HS-FPN: High frequency and spatial perception FPN for tiny object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 7, pp. 6896–6904.
[41] E. Ouyang, B. Li, W. Hu, G. Zhang, L. Zhao, and J. Wu, "When multigranularity meets spatial–spectral attention: A hybrid transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401118.
[42] N. Chen, L. Fang, Y. Xia, S. Xia, H. Liu, and J. Yue, "Spectral query spatial: Revisiting the role of center pixel in transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5402714.
[43] W.-T. Zhang, N. Xu, Y. Bai, and Y.-R. Zhang, "Multiscale sample transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5536418.



Zhenqiu Shu (Member, IEEE) received the Ph.D. degree in control science and engineering from Nanjing University of Science and Technology, Nanjing, China, in 2015.

He was a Visiting Scholar with the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia, from November 2014 to May 2015, under the supervision of Prof. Jun Zhou. He worked as a Postdoctoral Researcher with Jiangnan University for four years, under the guidance of Professor Xiao-Jun Wu. In February 2021, he joined the Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming, China, where he is currently a Full Professor. He received published more than 100 research papers in refereed international journals and conferences including AAAI, IJCAI, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON BIG DATA, IEEE/ACM TRANSACTIONS ON COMPUTATIONAL BIOLOGY AND BIOINFORMATICS, *ACM Transactions on Asian and Low-Resource Language Information Processing*, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, IEEE JOURNAL OF BIOMEDICAL AND HEALTH INFORMATICS, PR, KBS, INS, ASOC, ESWA, NeuNet, EAAI, INS, JCIM, APIN, and NEUCOM. His research interests include pattern recognition, computer vision, and machine learning.



Nanbo Liu is currently pursuing the M.S. degree in artificial intelligence with the Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming, China.

His research interests include remote sensing and computer vision.



Kexin Zeng is currently pursuing the M.S. degree with the Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming, China.

She received published several research works in high-impact journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS. Her research interests include remote sensing and computer vision, with a particular focus on hyperspectral image analysis.

Ms. Zeng was awarded the Graduate Elite Innovation Program in 2025 by Kunming University of Science and Technology.



Hongbin Wang received the Ph.D. degree in computer system architecture from Jilin University, Changchun, China, in 2013.

He is currently a Professor with the School of Information Engineering and Automation, Kunming University of Science and Technology, Kunming, China. His main research interests include natural language processing, data mining, and machine learning.



Jun Zhou (Fellow, IEEE) received the Ph.D. degree in computer science from the University of Alberta, Edmonton, AB, Canada, in 2006.

He was a Research Fellow with the Research School of Computer Science, The Australian National University, Canberra, ACT, Australia, and a Researcher with Canberra Research Laboratory, National Information and Communications Technology Australia, Canberra. In 2012, he joined the School of Information and Communication Technology, Griffith University, Nathan, QLD, Australia, where he is a Professor. His research interests include pattern recognition, computer vision, and spectral imaging and their applications in remote sensing and environmental informatics.

Dr. Zhou is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *Pattern Recognition*.

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